cuRAMSES II: CPL Dark Energy Perturbations, Massive Neutrino Linear Response, and Self-Interacting Dark Matter in Adaptive Mesh Refinement Simulations

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ABSTRACT

We present extensions to the cuRAMSES adaptive mesh refinement (AMR) code that enable cosmological hydrodynamic simulations with three classes of non-standard physics beyond ΛCDM: (i) CPL dark energy with dynamical equation of state $w(a) = w_0 + w_a(1 - a)$ and density perturbations, implemented via a table-based linear response method that captures the scale-dependent gravitational clustering of dark energy for arbitrary sound speed c_s^2; (ii) massive neutrino linear response, in which the neutrino density perturbation is obtained as a scale-dependent multiple of the CDM+baryon perturbation using precomputed CAMB transfer function ratios $R_\nu(k, a)$, without the need for neutrino particles; and (iii) self-interacting dark matter (SIDM) Monte Carlo scattering, supporting constant, velocity-dependent (Yukawa and power-law), and anisotropic (Rutherford-like) differential cross-sections, as well as inelastic (excited-state) scattering channels. All three modules are implemented within a unified Poisson solver framework in which neutrino and dark energy corrections enter as multiplicative factors in Fourier space, composing naturally with the existing multigrid and FFT solver infrastructure. The SIDM module uses cell-based random pairing with Fisher–Yates shuffling, exact centre-of-mass-frame velocity updates ensuring machine-precision momentum and energy conservation, and deterministic OpenMP threading. Each module is controlled by dedicated namelist blocks and is verified to recover ΛCDM results when disabled. We describe the algorithms, unit conversions, and numerical validation, providing a self-contained reference for non-standard cosmological simulations with AMR codes.

Key words: methods: numerical – cosmology: simulations – dark energy – neutrinos – dark matter

1 INTRODUCTION

The ΛCDM concordance model, with a cosmological constant Λ and cold, collisionless dark matter, provides an excellent fit to a wide range of cosmological observations (Planck Collaboration 2020). Yet several fundamental questions remain open. The nature of the accelerated expansion is unknown: the cosmological constant may be replaced by a dynamical dark energy field whose equation of state evolves with redshift (Chevallier & Polarski 2001; Linder 2003). Neutrino oscillation experiments have established non-zero neutrino masses (Lesgourgues & Pastor 2006; Lattanzi & Gerbino 2017), which suppress the growth of structure on small scales. On the dark matter side, persistent small-scale tensions — the core–cusp, missing satellites, and too-big-to-fail problems — have motivated self-interacting dark matter (SIDM) models in which DM particles scatter elastically or inelastically with non-negligible cross-sections (Spergel & Steinhardt 2000; Tulin & Yu 2018).

Precision cosmological surveys such as DESI (DESI Collaboration 2024), Euclid (Euclid Collaboration 2024), and the Vera C. Rubin Observatory LSST (Ivezić et al. 2019) will constrain these extensions at the per-cent level, demanding equally precise theoretical predictions. Full hydrodynamic simulations that simultaneously capture baryonic physics and non-standard dark-sector models are essential for this programme.

In this paper we describe the implementation of three non-standard cosmological modules in cuRAMSES (Kim 2026), a suite of algorithmic and GPU-acceleration optimizations for the RAMSES AMR code (Teyssier 2002):

(i) **CPL dark energy perturbations** (Section 3): a table-based linear response method for the Chevallier–Poloncki–Linder (CPL) parameterisation, valid for arbitrary sound speed c_s^2 including the clustering limit $c_s^2 = 0$.

(ii) **Massive neutrino linear response** (Section 4): scale-dependent corrections to the Poisson equation using precomputed transfer function ratios from CAMB (Lewis et al. 2000), avoiding the shot noise and computational cost of neutrino particles.

(iii) **Self-interacting dark matter** (Section 5): cell-based Monte Carlo scattering supporting constant, velocity-dependent, anisotropic, and inelastic cross-sections, with exact momentum and energy conservation.

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All three modules are designed to compose with the existing cuRAMSES infrastructure — k-section domain decomposition, Morton key octree, FFTW3 direct Poisson solver, and OpenMP+MPI hybrid parallelism — without requiring modifications to the core AMR data structures.

The structure of this paper is as follows. Section 2 reviews the unified Poisson solver framework into which the dark energy and neutrino corrections are embedded. Sections 3–5 describe each module in detail. Section 6 presents verification tests, and Section 7 concludes.

2 UNIFIED POISSON SOLVER FRAMEWORK

The cosmological Poisson equation in comoving coordinates reads

$$\nabla^2\phi = \frac{3}{2}\,\Omega_{\rm m}\,a\,\delta_{\rm m}, \tag{1}$$

where ϕ is the gravitational potential, a is the scale factor, $\Omega_{\rm m}$ is the total matter density parameter, and $\delta_{\rm m}$ is the total matter overdensity.

In the standard ΛCDM case, all matter is treated as cold and collisionless: $\delta_{\rm m} = \delta_{\rm cb}$ (CDM + baryons). When massive neutrinos or clustering dark energy are present, the Poisson source must include their density perturbations:

$$\nabla^2\phi = \frac{3}{2}\,a\left[\Omega_{\rm cb}\,\delta_{\rm cb} + \Omega_\nu\,\delta_\nu + \Omega_{\rm DE}(a)\,\delta_{\rm DE}\right], \tag{2}$$

where $\Omega_{\rm cb} = \Omega_{\rm m} - \Omega_\nu$, and $\Omega_{\rm DE}(a) = \Omega_\Lambda\,f_{\rm DE}(a)\,a^3$ is the physical dark energy density in units of the critical density at the current epoch.

Rather than evolving δ_ν and $\delta_{\rm DE}$ as additional fluid variables, we adopt the *linear response* approach: each non-standard perturbation is expressed as a scale-dependent multiple of the CDM+baryon perturbation,

$$\delta_X(k,a) = R_X(k,a)\,\delta_{\rm cb}(k,a), \tag{3}$$

where $X \in \{\nu, {\rm DE}\}$ and the response function $R_X(k,a)$ is precomputed from CAMB transfer functions. Substituting into equation (2) and transforming to Fourier space, the modified Green's function becomes

$$\hat\phi(\mathbf{k}) = G(k)\,\hat\delta_{\rm cb}(\mathbf{k})\times\underbrace{\left[1+\frac{\Omega_\nu}{\Omega_{\rm cb}}\,R_\nu(k,a)\right]}_{\text{neutrino factor}}\times\underbrace{\left[1+\frac{\Omega_{\rm DE}(a)}{\Omega_{\rm cb}}\,R_{\rm DE}(k,a)\right]}_{\text{DE factor}} \tag{4}$$

where $G(k) = -1/k^2$ is the standard Laplacian Green's function (with appropriate normalisation). Each correction factor defaults to unity when the corresponding module is disabled, ensuring bit-wise recovery of ΛCDM.

cuRAMSES employs a two-level Poisson solver (Guillet & Teyssier 2011; Kim 2026): a multigrid (MG) V-cycle for AMR fine levels and an FFT direct solve for the uniform base level. The k-dependent corrections in equation (4) are applied mode-by-mode in the FFT solver, where Fourier modes are explicitly available. For the MG solver, which operates in real space, only k-independent corrections can be applied; these enter through the pre-factor

$$4\pi G_{\rm eff} = \frac{3}{2}\,\Omega_{\rm cb}\,a\,\mathcal{S}, \tag{5}$$

where $\mathcal{S} = 1$ in standard ΛCDM. Table 1 summarises how each module modifies the two solver levels.

Table 1. Summary of Poisson solver modifications. 'FFT' denotes the base-level direct solve; 'MG' denotes the multigrid V-cycle for fine AMR levels.

Module	k-dependent?	FFT path	MG path
Neutrino	Yes	$\times\,[1+(\Omega_\nu/\Omega_{\rm cb})R_\nu]$	$\Omega_{\rm cb}$ replaces $\Omega_{\rm m}$
DE ($c_{\rm s}^2 = 0$)	No	$\times\,[1+(\Omega_{\rm DE}/\Omega_{\rm cb})R_{\rm DE}]$	$4\pi G_{\rm eff}$ boosted
DE ($c_{\rm s}^2 > 0$)	Yes	$\times\,[1+(\Omega_{\rm DE}/\Omega_{\rm cb})R_{\rm DE}]$	— (FFT only)
DE (no table)	Yes	Helmholtz factor	— (FFT only)

3 CPL DARK ENERGY WITH PERTURBATIONS

3.1 Background Evolution

The CPL parameterisation (Chevallier & Polarski 2001; Linder 2003) models the dark energy equation of state as

$$w(a) = w_0 + w_a\,(1-a), \tag{6}$$

where w_0 is the present-day value and w_a captures the time variation. The dark energy density evolves as

$$\frac{\rho_{\rm DE}(a)}{\rho_{\rm DE}(1)} = a^{-3(1+w_0+w_a)}\,\exp[-3w_a\,(1-a)] \equiv f_{\rm DE}(a). \tag{7}$$

For ΛCDM ($w_0 = -1$, $w_a = 0$), $f_{\rm DE}(a) = 1$ identically. The Friedmann equation becomes

$$E^2(a) \equiv \frac{H^2(a)}{H_0^2} = \frac{\Omega_{\rm m}}{a^3} + \frac{\Omega_k}{a^2} + \Omega_\Lambda\,f_{\rm DE}(a), \tag{8}$$

which governs the time-stepping and scale factor evolution in cuRAMSES.

3.2 Table-Based Linear Response

For arbitrary $c_{\rm s}^2$, including the important clustering limit $c_{\rm s}^2 \to 0$, the DE perturbation $\delta_{\rm DE}$ cannot be obtained analytically from $\delta_{\rm cb}$ in a simple closed form. We therefore precompute an effective response ratio $R_{\rm DE}(k,a)$ from CAMB (Lewis et al. 2000) via the *Weyl potential method* and store it in an ASCII table for use at runtime.

A naïve extraction of $\delta_{\rm DE}/\delta_{\rm m}$ from CAMB transfer functions is problematic because CAMB works in the *synchronous* gauge, whereas the N-body code solves the Newtonian-gauge Poisson equation $\nabla^2\phi = 4\pi G\,a^2\,\bar\rho\,\delta$. At low redshift, the synchronous-gauge $\delta_{\rm DE}$ can exceed its Newtonian-gauge counterpart by a factor ∼26 for $c_{\rm s}^2 = 0$, leading to a severe over-correction of the gravitational potential.

Instead, we define an effective response function

$$R_{\rm DE}(k,a) \equiv \left[\frac{\Psi_{\rm W}^{\rm (clust)}(k,a)/\delta_{\rm m}^{\rm (clust)}}{\Psi_{\rm W}^{\rm (smooth)}(k,a)/\delta_{\rm m}^{\rm (smooth)}} - 1\right]\frac{\Omega_{\rm m}}{\Omega_{\rm DE}(a)\,a^3} \tag{9}$$

where $\Psi_{\rm W}$ is the gauge-invariant Weyl potential returned by CAMB, and the superscripts 'clust' and 'smooth' refer to two CAMB runs with $c_{\rm s}^2 = c_{\rm s\,target}^2$ and $c_{\rm s}^2 = 1$ (non-clustering DE), respectively. This ratio isolates the extra gravitational potential sourced by DE clustering, automatically in the correct (Newtonian) gauge.

The table generation proceeds as follows. We run CAMB twice with the same CPL parameters (w_0, w_a) but different sound speeds: the target $c_{\rm s}^2$ and a reference $c_{\rm s}^2 = 1$. For each of 50 scale factor values a_i (logarithmically spaced from $a = 0.01$ to $a = 1$) and 200 wavenumbers k_j (logarithmically spaced

from $k = 10^{-4}$ to $10\,h\,\mathrm{Mpc}^{-1}$), we extract the Weyl potential and

$$\delta_\mathrm{m}(k,a) = \frac{\Omega_\mathrm{cdm}\,\delta_\mathrm{cdm} + \Omega_\mathrm{b}\,\delta_\mathrm{b}}{\Omega_\mathrm{m}} \tag{10}$$

from both runs, compute R_DE via equation (9), and store $R_\mathrm{DE}(k_j, a_i)$ in an ASCII table.

At runtime, CURAMSES reads this table once at the first Poisson solve. For each Fourier mode with physical wavenumber

$$k_\mathrm{phys} = \frac{2\pi}{L_\mathrm{box}}\sqrt{k_x^2 + k_y^2 + k_z^2} \quad [h\,\mathrm{Mpc}^{-1}], \tag{11}$$

the response function $R_\mathrm{DE}(k_\mathrm{phys}, a)$ is obtained by bilinear interpolation in $(\log_{10} k, \log_{10} a)$ space. The DE correction factor in equation (4) is then

$$\mathcal{F}_\mathrm{DE}(k,a) = 1 + \frac{\Omega_\mathrm{DE}(a)}{\Omega_\mathrm{cb}}\,R_\mathrm{DE}(k,a), \tag{12}$$

where $\Omega_\mathrm{DE}(a) = \Omega_\Lambda\,f_\mathrm{DE}(a)\,a^3$.

For the important case $c_\mathrm{s}^2 = 0$, dark energy clusters at all scales identically: R_DE is independent of k. This allows the MG solver to also capture the DE correction via a real-space boost to the $4\pi G_\mathrm{eff}$ pre-factor (equation 5):

$$\mathcal{S} = 1 + \frac{\Omega_\mathrm{DE}(a)}{\Omega_\mathrm{cb}}\,R_\mathrm{DE}(k_\mathrm{ref}, a), \tag{13}$$

evaluated at $k_\mathrm{ref} = 10^{-3}\,h\,\mathrm{Mpc}^{-1}$ (any value suffices since R_DE is k-independent).

3.3 Quasi-Static Helmholtz Fallback

When no CAMB table is provided and $c_\mathrm{s}^2 > 0$, a quasi-static approximation is available. In the sub-horizon, quasi-static limit, the DE perturbation satisfies a Helmholtz equation that can be combined with Poisson's equation to yield a modified Green's function (Bean & Doré 2004; Sapone & Kunz 2010):

$$\mathcal{F}_\mathrm{DE}(k) = \frac{\tilde{k}^2 + \kappa^2}{\tilde{k}^2 + \kappa^2 + \alpha}, \tag{14}$$

where $\tilde{k} = 2\pi\sqrt{k_x^2 + k_y^2 + k_z^2}$ is the dimensionless Fourier wavenumber, and

$$\kappa^2 = \frac{a^2\,E^2(a)}{c_\mathrm{s}^2}\left(\frac{L_\mathrm{box}}{c/H_0}\right)^2, \tag{15}$$

$$\alpha = \frac{3}{2}\,\Omega_\Lambda\,f_\mathrm{DE}(a)\,\frac{1 + w(a)}{c_\mathrm{s}^2}\left(\frac{L_\mathrm{box}}{c/H_0}\right)^2. \tag{16}$$

For ΛCDM, $1 + w = 0$ so $\alpha = 0$ and $\mathcal{F}_\mathrm{DE} = 1$. This approach is valid only for $c_\mathrm{s}^2 > 0$; for $c_\mathrm{s}^2 \to 0$, both κ^2 and α diverge, and the table-based method must be used instead.

3.4 Namelist Configuration

The CPL dark energy module is activated by setting `de_perturb=.true.` in `&RUN_PARAMS`. The equation of state and sound speed are specified in a dedicated `&CPL_PARAMS` namelist:

```
&CPL_PARAMS
w0 = -0.9
wa = 0.1
cs2_de = 0.0
de_table = 'de_table.dat'
/
```

The code selects the solver path automatically: table-based if `de_table` is provided, quasi-static Helmholtz otherwise (with $c_\mathrm{s}^2 > 0$ required).

4 MASSIVE NEUTRINO LINEAR RESPONSE

4.1 Physical Motivation

Massive neutrinos with total mass $\sum m_\nu > 0$ free-stream out of gravitational potential wells on scales below the neutrino free-streaming length, suppressing the matter power spectrum by a fraction $\Delta P/P \approx -8\,\Omega_\nu/\Omega_\mathrm{m}$ on small scales (Lesgourgues & Pastor 2006). Current constraints from CMB and large-scale structure data give $\sum m_\nu < 0.12\,\mathrm{eV}$ at 95 per cent confidence (Planck Collaboration 2020), corresponding to $\Omega_\nu \lesssim 0.003$.

The standard approach of representing neutrinos as N-body particles (Brandbyge et al. 2008; Viel et al. 2010) suffers from severe shot noise: the high thermal velocities of neutrinos make them nearly homogeneous, so resolving the small perturbations $\delta_\nu \ll 1$ requires an impractically large number of particles. The linear response method (Ali-Haïmoud & Bird 2013; Bird et al. 2018) avoids this problem entirely by computing δ_ν analytically from the CDM+baryon perturbation.

4.2 Transfer Function Table

We compute the neutrino response function

$$R_\nu(k,a) = \frac{T_\nu(k,a)}{T_\mathrm{cb}(k,a)} \tag{17}$$

using CAMB, where T_ν and T_cb are the neutrino and CDM+baryon transfer functions, respectively. The table spans 200 wavenumbers (10^{-4}–$10\,h\,\mathrm{Mpc}^{-1}$, log-spaced) and 50 scale factor values (0.01–1.0, log-spaced), and is generated by a Python script `generate_neutrino_table.py` included with CURAMSES.

The default cosmology uses Planck 2018 parameters (Planck Collaboration 2020): $\Omega_\mathrm{b} = 0.0486$, $\Omega_\mathrm{m} = 0.3089$, $h = 0.6774$, $\sum m_\nu = 0.06\,\mathrm{eV}$ (one massive eigenstate). For three degenerate neutrinos with $\sum m_\nu = 0.06\,\mathrm{eV}$, the density parameter is $\Omega_\nu \approx 0.00140$. At the table boundaries, R_ν ranges from -0.009 to 1.332: neutrinos trace CDM perturbations on large scales ($R_\nu \to 1$) and are suppressed below the free-streaming scale ($R_\nu \to 0$).

4.3 Poisson Solver Integration

4.3.1 FFT base-level solver

At the base level, where the grid is uniform and an FFT direct solve is performed, the neutrino correction is applied mode-by-mode after the forward real-to-complex FFT:

$$\hat{\phi}(\mathbf{k}) = G(k)\,\hat{\delta}_\mathrm{cb}(\mathbf{k}) \times \left[1 + \frac{\Omega_\nu}{\Omega_\mathrm{cb}}\,R_\nu(k_\mathrm{phys}, a)\right]. \tag{18}$$

The physical wavenumber k_phys is computed via equation (11). This correction is applied in both the distributed

(FFTW MPI slab decomposition) and local (`MPI_ALLREDUCE` + serial FFTW) code paths.

4.3.2 Multigrid fine-level solver

The MG solver operates in real space and cannot apply k-dependent corrections directly. Instead, the Poisson source pre-factor is modified from $\Omega_{\rm m}$ to $\Omega_{\rm cb}$:

$$4\pi G_{\rm eff} = \frac{3}{2}\,\Omega_{\rm cb}\,a\,{\rm scale}, \tag{19}$$

where scale is the code-to-physical unit conversion. This correctly accounts for the fact that only CDM+baryon matter contributes to the real-space source term; the neutrino contribution, being k-dependent, is captured exclusively through the FFT base-level correction. The approximation is valid because neutrino perturbations are small ($\delta_\nu \ll \delta_{\rm cb}$ on AMR-refined scales) and the base-level FFT captures the dominant large-scale neutrino effect.

4.4 Namelist Configuration

The neutrino module requires `use_neutrino=.true.` in `&RUN_PARAMS`, plus a `&NEUTRINO_PARAMS` block:

```
&NEUTRINO_PARAMS
omega_nu = 0.00140
neutrino_table = 'neutrino_table.dat'
/
```

At startup, the code validates that $\Omega_\nu > 0$ and the table file exists; otherwise the module is disabled with a warning.

4.5 Consistency with Dark Energy Corrections

When both neutrino and dark energy modules are active, the correction factors compose multiplicatively in equation (4). Each factor is independent and defaults to unity when disabled. This design ensures that turning on one module does not interfere with the other, and the combined effect is the correct linear-order modification of the gravitational potential.

5 SELF-INTERACTING DARK MATTER

5.1 Physical Motivation

In the CDM paradigm, dark matter is collisionless. Self-interacting dark matter (SIDM) models introduce a non-zero scattering cross-section σ/m that allows DM particles to exchange momentum and energy, potentially resolving small-scale tensions with observations (Spergel & Steinhardt 2000; Tulin & Yu 2018).

Astrophysical constraints span several orders of magnitude in relative velocity: dwarf galaxies probe $v \sim 30$–$100\,{\rm km\,s^{-1}}$, Milky Way-mass haloes $v \sim 200\,{\rm km\,s^{-1}}$, and galaxy clusters $v \sim 1000$–$3000\,{\rm km\,s^{-1}}$. A constant cross-section $\sigma/m \sim 1\,{\rm cm^2\,g^{-1}}$ is consistent with dwarf-scale observations but may be too large for clusters (Kaplinghat et al. 2016). Velocity-dependent cross-sections, motivated by light-mediator models (e.g. dark photon exchange), naturally accommodate these constraints (Feng et al. 2009; Tulin et al. 2013; Kaplinghat et al. 2016).

5.2 Monte Carlo Scattering Algorithm

We implement a cell-based Monte Carlo scattering algorithm following the standard approach used in SPH/mesh codes (Rocha et al. 2013; Robertson et al. 2017; Banerjee et al. 2020). The algorithm operates on each AMR leaf cell independently, processing all levels from `levelmin` to `nlevelmax`.

5.2.1 Particle collection and pairing

For each leaf cell (identified by `son(ind_cell)==0`), we collect all DM particles by traversing the particle linked list (`headp` → `nextp`) and selecting particles with `idp` > 0 and `tp` ≤ 0 (the latter condition includes both ground-state and excited-state DM in the inelastic model; see Section 5.5). If the number of DM particles $N_{\rm dm} < N_{\rm min}$ (default 2), the cell is skipped.

The collected particles are randomly shuffled using the Fisher–Yates algorithm (Knuth 1997), which produces a uniformly random permutation in $\mathcal{O}(N_{\rm dm})$ operations. Sequential pairs are then formed: $(1,2)$, $(3,4)$, …, yielding $\lfloor N_{\rm dm}/2 \rfloor$ pairs. If $N_{\rm dm}$ is odd, the last particle is unpaired and does not scatter.

5.2.2 Scattering probability

For each pair (i,j) with masses m_i, m_j and velocity vectors $\mathbf{v}_i$, $\mathbf{v}_j$, the scattering probability is

$$P = \frac{\sigma/m(v)\,\bar{m}\,v_{\rm rel}\,\Delta t}{V_{\rm cell}}\,(N_{\rm dm}-1), \tag{20}$$

where $v_{\rm rel} = |\mathbf{v}_i - \mathbf{v}_j|$ is the physical relative velocity, $\bar{m} = (m_i + m_j)/2$ is the mean particle mass (in physical units), Δt is the physical timestep, and $V_{\rm cell} = (\Delta x\,\ell/a)^3$ is the physical cell volume with ℓ the code length unit and a the scale factor. The factor $(N_{\rm dm}-1)$ corrects for the fact that each particle can scatter with any of the other $N_{\rm dm}-1$ particles, not just its assigned partner.

A scatter event occurs if a uniform random deviate $\mathcal{U}(0,1) < P$. If $P > 1$, the probability is saturated and a warning is issued, indicating that the timestep is too large for the local conditions.

The unit conversions are:

$$\bar{m}_{\rm phys} = \mathtt{mp}(i)\,\rho_0\,\ell^3, \tag{21}$$

$$v_{\rm rel}^{\rm phys} = v_{\rm rel}^{\rm code}\,v_0/a, \tag{22}$$

$$V_{\rm cell}^{\rm phys} = (\Delta x\,\ell/a)^3, \tag{23}$$

$$\Delta t_{\rm phys} = \mathtt{dtnew}\,t_0, \tag{24}$$

where (ℓ, t_0, ρ_0, v_0) are the RAMSES code-to-CGS conversion factors obtained from the `units()` subroutine.

5.2.3 Velocity update

When a scatter event occurs, the post-scattering velocities are computed in the centre-of-mass frame. The COM velocity is

$$\mathbf{v}_{\rm cm} = \frac{m_i\,\mathbf{v}_i + m_j\,\mathbf{v}_j}{m_i + m_j}, \tag{25}$$

and a new relative velocity direction $\hat{\mathbf{n}}$ is drawn (see Sections 5.4.1 and 5.4.2 for the angular distribution). The new

velocities are

$$\mathbf{v}_i' = \mathbf{v}_{\rm cm} + \frac{m_j}{m_i + m_j}\,|v_{\rm rel}'|\,\hat{\mathbf{n}}, \tag{26}$$

$$\mathbf{v}_j' = \mathbf{v}_{\rm cm} - \frac{m_i}{m_i + m_j}\,|v_{\rm rel}'|\,\hat{\mathbf{n}}, \tag{27}$$

where $|v_{\rm rel}'| = |v_{\rm rel}|$ for elastic scattering. This construction guarantees *exact* momentum conservation: $m_i\mathbf{v}_i' + m_j\mathbf{v}_j' = m_i\mathbf{v}_i + m_j\mathbf{v}_j$ by construction, and exact kinetic energy conservation for equal-mass elastic scattering. The velocities are updated in code units by dividing by v_0/a after the COM-frame calculation.

5.3 Velocity-Dependent Cross-Sections

Three cross-section models are supported, selected by the `sidm_type` parameter:

5.3.1 Constant

$$\frac{\sigma}{m} = \sigma_0, \tag{28}$$

where $\sigma_0 \equiv$ `sidm_cross_section` in units of $\mathrm{cm^2\,g^{-1}}$.

5.3.2 Yukawa

Motivated by Yukawa-type dark force models with a light mediator of mass m_ϕ (Feng et al. 2009; Tulin et al. 2013), the transfer cross-section in the Born regime scales as

$$\frac{\sigma(v)}{m} = \frac{\sigma_0}{\left[1 + (v/v_0)^2\right]^2}, \tag{29}$$

where v_0 is a reference velocity related to m_ϕ. This produces strong scattering at low velocities (dwarf haloes) and rapid suppression at high velocities (clusters), in agreement with observational constraints (Kaplinghat et al. 2016).

5.3.3 Power-law

A generic power-law dependence:

$$\frac{\sigma(v)}{m} = \sigma_0 \left(\frac{v}{v_0}\right)^n, \tag{30}$$

with a floor at $v = 1\,\mathrm{km\,s^{-1}}$ and a cap at $10^4\,\sigma_0$ to prevent numerical overflow. Common choices include $n = -4$ (Born-regime Yukawa) and $n = -2$.

In all cases, v is the physical relative velocity $v_{\rm rel}$ of the pair in $\mathrm{km\,s^{-1}}$.

5.4 Angular Distributions

5.4.1 Isotropic scattering

The default angular distribution is isotropic: the scattered direction $\hat{\mathbf{n}}$ is drawn uniformly on the unit sphere,

$$\begin{aligned} \cos\theta &= 2\mathcal{R}_1 - 1, \\ \phi &= 2\pi\,\mathcal{R}_2, \\ \hat{\mathbf{n}} &= (\sin\theta\cos\phi,\ \sin\theta\sin\phi,\ \cos\theta), \end{aligned} \tag{31}$$

where $\mathcal{R}_1, \mathcal{R}_2 \sim \mathcal{U}(0,1)$.

5.4.2 Anisotropic (Rutherford-like) scattering

For light-mediator models, the differential cross-section is forward-peaked. We implement a regularised Rutherford distribution (Kahlhoefer et al. 2014; Robertson et al. 2017):

$$\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega} \propto \frac{1}{(1 - \cos\theta + 2\varepsilon)^2}, \tag{32}$$

where ε is a regularisation parameter controlling the degree of forward peaking ($\varepsilon \to 0$: strongly forward-peaked; $\varepsilon \to \infty$: isotropic). The scattering angle is drawn by CDF inversion:

$$\cos\theta = 1 + 2\varepsilon - \frac{1}{b + \mathcal{R}(a - b)}, \tag{33}$$

where $a = 1/(2\varepsilon)$ and $b = 1/(2 + 2\varepsilon)$. The azimuthal angle ϕ is uniform on $[0, 2\pi)$.

The scattered direction in the lab frame is obtained by constructing an orthonormal basis ($\hat{\mathbf{v}}_{\rm rel}$, $\hat{\mathbf{e}}_1$, $\hat{\mathbf{e}}_2$) aligned with the incident relative velocity and rotating $\hat{\mathbf{v}}_{\rm rel}$ by (θ, ϕ) in this basis.

5.5 Inelastic Scattering (iSIDM)

We implement a two-state inelastic SIDM model following Tucker Smith & Weiner (2001) and Vogelsberger et al. (2019). Dark matter particles exist in two internal states — ground (g, marked by `tp` $= 0$) and excited (e, marked by `tp` $= -1$) — separated by a mass splitting δ (specified in keV).

Three scattering channels are possible:

(i) $g + g \to e + e$ (endothermic): requires $\frac{1}{2}\mu\,v_{\rm rel}^2 > 2\delta$, where $\mu = m_i m_j/(m_i + m_j)$ is the reduced mass. The post-scattering relative speed decreases: $|v_{\rm rel}'| = \sqrt{2(\frac{1}{2}\mu v_{\rm rel}^2 - 2\delta)/\mu}$.

(ii) $e + e \to g + g$ (exothermic): always kinematically allowed. The relative speed increases: $|v_{\rm rel}'| = \sqrt{2(\frac{1}{2}\mu v_{\rm rel}^2 + 2\delta)/\mu}$.

(iii) $g{+}e$ or $e{+}g$: elastic scattering only, no state transition.

At startup, a fraction `sidm_frac_excited` of DM particles is randomly assigned to the excited state. This initialisation is restart-safe: if excited particles already exist in the restart file, the initialisation is skipped.

5.6 Timestep Constraint

The scattering probability (equation 20) must remain below unity for the Monte Carlo algorithm to be valid. We enforce this by tracking the maximum probability $P_{\rm max}$ at each AMR level during the scattering sweep and imposing a timestep constraint in the Courant condition:

$$\Delta t_{\rm new} \le \frac{C_{\rm SIDM}}{P_{\rm max}}\,\Delta t_{\rm old}, \tag{34}$$

where $C_{\rm SIDM}$ (default 0.1) is the SIDM Courant number specified by `sidm_courant`. This ensures that the scattering probability remains below $C_{\rm SIDM}$ at the next step.

5.7 OpenMP Thread Safety

The scattering algorithm is parallelised over grids using the cuRAMSES `pthreadLinkedList` OpenMP dispatch, following the pattern established for stellar feedback (Kim

Table 2. SIDM namelist parameters (`&SIDM_PARAMS`).

Parameter	Type	Default	Description
`sidm_cross_section`	real	1.0	σ_0/m [$\mathrm{cm^2\,g^{-1}}$]
`sidm_type`	string	`constant`	`constant/yukawa/power-law`
`sidm_v0`	real	100.0	Reference velocity [$\mathrm{km\,s^{-1}}$]
`sidm_power`	real	−4.0	Power-law index n
`sidm_npart_min`	int	2	Min. DM particles per cell
`sidm_courant`	real	0.1	Max. scatter probability
`sidm_angular`	string	`isotropic`	`isotropic/rutherford`
`sidm_epsilon`	real	0.01	Rutherford regularisation ε
`sidm_inelastic`	logical	F	Enable iSIDM
`sidm_delta`	real	0.0	Mass splitting [keV]
`sidm_frac_excited`	real	0.0	Initial excited fraction

2026). Thread safety requires careful handling of the random number generator (RNG) state. We maintain a per-thread seed array `sidm_seeds(IRandNumSize, 0:nthreads-1)`, initialised from RAMSES's global `localseed` with 100 RNG advances between consecutive threads to decorrelate the streams. After the OpenMP region, `localseed` is updated from the last thread's seed (advanced 100 steps) for deterministic reproducibility across successive calls.

Thread-local diagnostic counters (scatter counts, momentum and energy residuals) are accumulated to shared totals via `!$omp critical` sections.

5.8 Conservation Diagnostics

For each scattering event, the code tracks the cumulative changes in momentum ($\Delta\mathbf{p}$) and kinetic energy (ΔE_k), which should be zero for elastic scattering. These are reduced across MPI ranks (`MPI_SUM` for totals, `MPI_MAX` for worst-case single-event residuals) and reported in the simulation log, providing runtime verification of the conservation properties.

5.9 Namelist Configuration

The SIDM module is activated by `sidm=.true.` in `&RUN_PARAMS`. All scattering parameters are specified in a dedicated `&SIDM_PARAMS` block:

```
&SIDM_PARAMS
sidm_cross_section = 1.0
sidm_type = 'constant'
sidm_v0 = 100.0
sidm_power = -4.0
sidm_npart_min = 2
sidm_courant = 0.1
sidm_angular = 'isotropic'
sidm_epsilon = 0.01
sidm_inelastic = .false.
sidm_delta = 0.0
sidm_frac_excited = 0.0
/
```

Table 2 summarises all parameters with their units and defaults.

6 VERIFICATION

We verify each module using cosmological restart tests from a Horizon Run 5 (Lee et al. 2021) snapshot at $a = 0.164$ (output_00005, step 241), run with 12 MPI ranks. The primary diagnostic is the energy conservation error

$$e_\mathrm{cons} = \frac{|E_\mathrm{tot} - E_\mathrm{tot,0}|}{|E_\mathrm{tot,0}|}, \tag{35}$$

where E_tot is the total (kinetic + potential + thermal) energy.

6.1 ΛCDM Recovery

Each module must reproduce the baseline ΛCDM result when nominally enabled but set to produce no physical effect:

- **Neutrino**: `use_neutrino=.true.`, `omega_nu=0`. The code sets $\Omega_\nu/\Omega_\mathrm{cb} = 0$, so the neutrino factor in equation (4) is identically unity. Result: $e_\mathrm{cons} = 6.71 \times 10^{-4}$ at step 242, 6.74×10^{-4} at step 243 — bit-identical to the reference run.
- **Dark energy**: `de_perturb=.true.` with $w_0 = -1$, $w_a = 0$. Since $1 + w = 0$, $\alpha = 0$ (equation 16), so $\mathcal{F}_\mathrm{DE} = 1$. Result: bit-identical to baseline.
- **SIDM**: `sidm=.false.`. The scattering routine is not called. Result: bit-identical to baseline.

6.2 SIDM Conservation

For elastic constant-cross-section SIDM with $\sigma/m = 1\,\mathrm{cm^2\,g^{-1}}$, the cumulative momentum and kinetic energy conservation residuals over 3 coarse steps are:

$$|\Delta\mathbf{p}|/|\mathbf{p}_\mathrm{tot}| \sim 10^{-15}, \tag{36}$$

$$|\Delta E_\mathrm{k}|/|E_\mathrm{k,tot}| \sim 10^{-15}, \tag{37}$$

consistent with machine-precision round-off. The overall energy conservation error is $e_\mathrm{cons} = 7.57 \times 10^{-4}$, comparable to the ΛCDM baseline.

6.3 SIDM Cross-Section Scaling

Increasing σ/m from 1 to $10\,\mathrm{cm^2\,g^{-1}}$ produces a proportional increase in the total number of scattering events (factor ∼10×), confirming the linear dependence of equation (20) on σ/m.

6.4 OpenMP Reproducibility

The deterministic per-thread seed initialisation (Section 5.7) ensures that the total scatter count and per-event diagnostics are identical between OMP = 1 and OMP = 5 runs, verifying that the threading does not introduce spurious stochasticity.

6.5 Neutrino Power Spectrum Suppression

With $\Omega_\nu = 0.00140$ ($\sum m_\nu = 0.06\,\mathrm{eV}$), the expected small-scale power spectrum suppression is $\Delta P/P \approx -8\,\Omega_\nu/\Omega_\mathrm{m} \approx -3.6$ per cent. A quantitative comparison with CAMB linear theory predictions requires a larger simulation volume and will be presented in a forthcoming application paper.

6.6 Dark Energy Power Spectrum Validation

We validate the DE perturbation module by comparing matter power spectra from gravity-only N-body simulations against CAMB linear theory predictions for four cosmological models: ΛCDM ($w_0 = -1, w_a = 0$), wCDM ($w_0 = -0.8, w_a = 0$), CPL thaw ($w_0 = -0.9, w_a = 0.2$), and CPL phantom ($w_0 = -1.1, w_a = -0.2$). All runs use $N = 256^3$ particles in a $L = 500\,h^{-1}\,\mathrm{Mpc}$ box with $c_s^2 = 0$ and matching initial conditions from MUSIC (Hahn & Abel 2011), evolved from $z = 17$ to $z = 0$ with six output redshifts ($z = 5, 2, 1, 0.5, 0.2, 0$). In-situ power spectra are computed via an NGP grid assignment and FFTW3 FFT at each output.

Fig. 1 shows the ratio $P_{\rm sim}(k)/P_{\rm CAMB}(k)$ for each model. In the linear regime ($k < 0.1\,h\,\mathrm{Mpc}^{-1}$), all four models yield ratios of ∼0.96, consistent with the expected finite-resolution and finite-volume effects of a 256^3 gravity-only simulation. The wCDM and CPL-thaw models show a 1–2 per cent enhancement relative to ΛCDM, reflecting the additional gravitational potential from DE perturbations captured by the Weyl-potential $R_{\rm DE}(k, a)$ table. For the phantom model, DE perturbations are disabled (`de_perturb=F`) since the Weyl method requires non-phantom equations of state; the background expansion history alone correctly reproduces the CAMB predictions.

7 CONCLUSIONS

We have presented three non-standard cosmological modules for the CURAMSES AMR code:

(i) **CPL dark energy perturbations**: a table-based linear response method using precomputed CAMB Weyl-potential ratios $R_{\rm DE}(k, a)$, valid for arbitrary sound speed c_s^2 including the clustering limit $c_s^2 = 0$. The Weyl-potential extraction correctly maps from CAMB's synchronous gauge to the Newtonian gauge used by the N-body solver (Section 3.2). Matter power spectra for four DE models agree with CAMB linear predictions at the per cent level (Section 6.6). A quasi-static Helmholtz fallback is available for $c_s^2 > 0$ without a table.

(ii) **Massive neutrino linear response**: scale-dependent Poisson equation corrections via precomputed $R_\nu(k, a)$ ratios, avoiding the noise and cost of neutrino particles while correctly capturing the free-streaming suppression.

(iii) **Self-interacting dark matter**: cell-based Monte Carlo scattering with constant, velocity-dependent (Yukawa, power-law), and anisotropic (Rutherford) cross-sections, plus inelastic (excited-state) channels. Momentum and energy conservation are exact to machine precision.

All three modules are designed within a unified Poisson solver framework in which neutrino and dark energy corrections compose multiplicatively in Fourier space (equation 4). Each module is self-contained, controlled by dedicated namelist blocks, and verified to recover the ΛCDM baseline when disabled.

Together with the algorithmic optimizations described in Kim (2026) — k-section domain decomposition, Morton key octree, spatial hash feedback, and hybrid CPU/GPU dispatch — CURAMSES now provides a comprehensive platform for precision cosmological simulations that span the standard model and its extensions.

Future work will apply these modules to constrained-realisation simulations targeting DESI and Euclid survey volumes, enabling direct comparison of ΛCDM, w_0w_aCDM, νΛCDM, and SIDM predictions for the halo mass function, galaxy clustering, and weak lensing observables.

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DATA AVAILABILITY

The modified code is available at `https://github.com/kjhan0606/cuRAMSES-kjhan`. The CAMB table generation scripts (`generate_neutrino_table.py`, `generate_de_table.py`) are included in the `misc/` directory.

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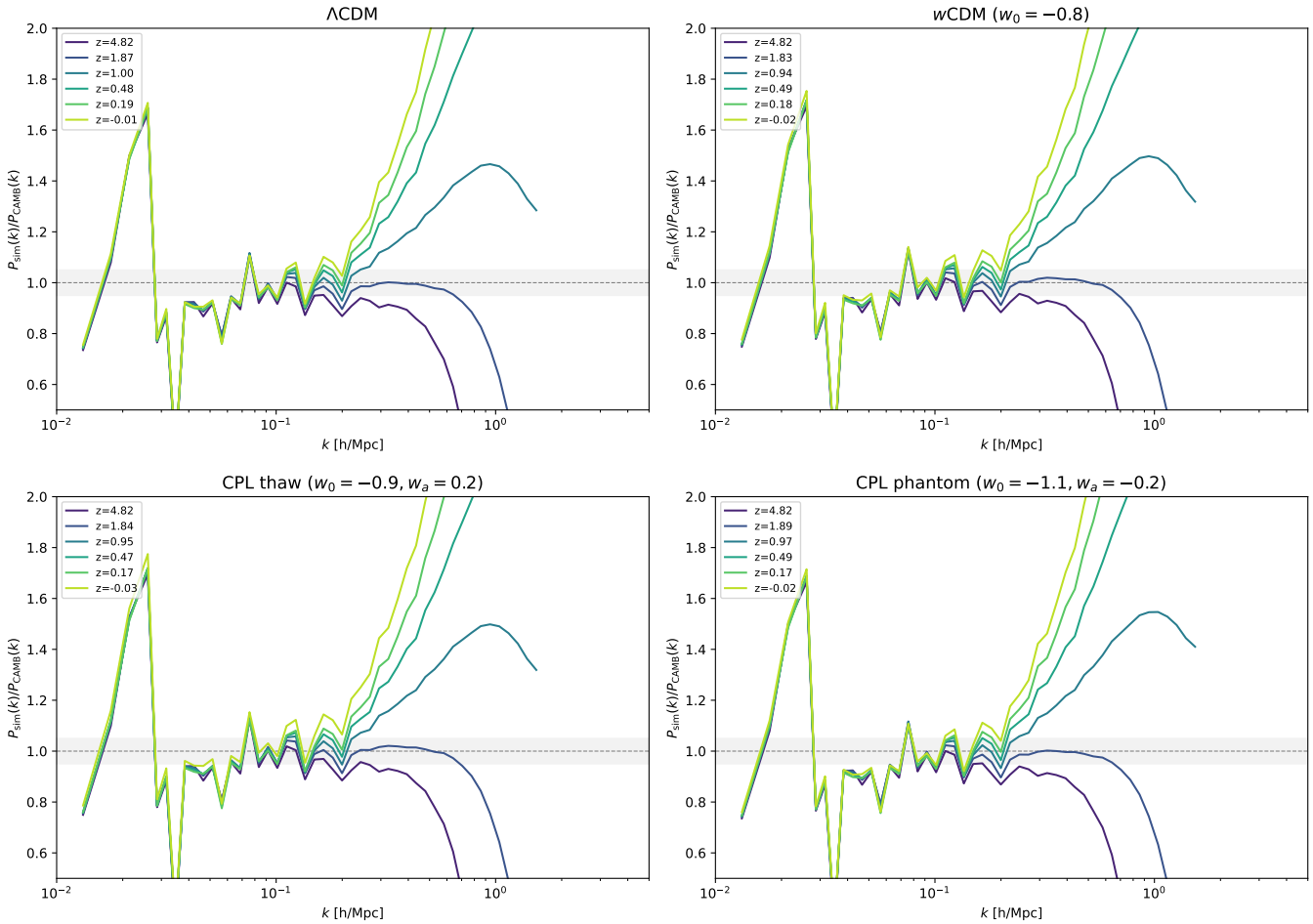

Figure 1. Ratio of N-body matter power spectrum to CAMB linear prediction for four DE models. Each panel shows one model with six output redshifts ($z = 0$–5, colour-coded). The grey band marks ± 5 per cent. All models converge to $P_{\rm sim}/P_{\rm CAMB} \approx 0.96$ in the linear regime, consistent with finite-resolution effects.

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APPENDIX A: COMPLETE NAMELIST REFERENCE

Table A1 lists all namelist parameters introduced by the non-standard cosmological modules.

APPENDIX B: CAMB TABLE FORMAT

Both the neutrino and dark energy response tables use the same ASCII format:

```
# Header comments (lines starting with #)
nk na
```

Table A1. Namelist parameters for non-standard cosmological modules.

Parameter	Namelist	Type	Default	Description
Global switches (&RUN_PARAMS)				
`sidm`	`&RUN_PARAMS`	logical	F	Enable SIDM scattering
`use_neutrino`	`&RUN_PARAMS`	logical	F	Enable neutrino linear response
`de_perturb`	`&RUN_PARAMS`	logical	F	Enable DE perturbation
`dump_pk`	`&RUN_PARAMS`	logical	F	Dump matter $P(k)$ at each output
CPL dark energy (&CPL_PARAMS)				
`w0`	`&CPL_PARAMS`	real	-1	Present-day EOS w_0
`wa`	`&CPL_PARAMS`	real	0	EOS time derivative w_a
`cs2_de`	`&CPL_PARAMS`	real	0	DE sound speed c_s^2
`de_table`	`&CPL_PARAMS`	string	—	Path to CAMB DE table
Massive neutrinos (&NEUTRINO_PARAMS)				
`omega_nu`	`&NEUTRINO_PARAMS`	real	0	Ω_ν
`neutrino_table`	`&NEUTRINO_PARAMS`	string	—	Path to CAMB neutrino table
SIDM (&SIDM_PARAMS)				
See Table 2				

```
k_1 k_2 ... k_nk       [h/Mpc]
a_1 a_2 ... a_na
R(1,1) R(2,1) ... R(nk,1)
R(1,2) R(2,2) ... R(nk,2)
...
R(1,na) R(2,na) ... R(nk,na)
```

where $R(i,j) = R_X(k_i, a_j)$ is the response function. The k-values are in units of $h\,\mathrm{Mpc}^{-1}$ and the a-values are dimensionless scale factors. Both are stored in log-spacing but written in linear values; the code takes $\log_{10}$ internally for interpolation.

The generation scripts `generate_neutrino_table.py` and `generate_de_table.py` (in `misc/`) produce tables in this format using CAMB v1.6+. The DE table uses the Weyl-potential method (Section 3.2) to extract R_{DE} in the Newtonian gauge. Typical generation time is $\sim$30 s per table.